Recitation (2)

Problem 2.1:

For scalar fields V and U,

a) Prove that $\nabla(UV) = U\nabla V + V\nabla U$
b) Verify part (a) by assuming that $V = 5xy$ and $U = y^2z$.

Problem 2.2:

1. Calculate the divergence of the given vectors at the specified points
   a) $\vec{E} = (1/z^2)[10xyz\hat{x} + 5x^2z\hat{y} + (2z^3 - 5x^2y)\hat{z}]$ at $P(-2,3,5)$;
   b) $\vec{D} = 5z^2\hat{\rho} + 10\rho z\hat{z}$ at $P(3, -45°, 5)$;
   c) $\vec{V} = 2r\sin\theta\sin\phi\hat{r} + r\cos\theta\sin\phi\hat{\theta} + r\cos\phi\hat{\phi}$ at $P(3, 45°, -45°)$.
2. Find the curl of the following vectors
   a) $\vec{H} = 2xz\hat{x} + 5xyz\hat{y} + 8(y + z)\hat{z}$
   b) $\vec{B} = r^2\hat{r} + 4r\cos 2\theta\hat{\theta}$

Problem 2.3:

Let $\vec{G} = 3xy^2\hat{x} + 2z\hat{y}$ Given an initial point $P(2,1,1)$ and a final point $Q(4,3,1)$, find $\int \vec{G} \cdot \overrightarrow{dl}$ using the path (a) straight line: $y = x - 1, z = 1$;

(b) parabola: $6y = x^2 + 2, z = 1$.

Problem 2.4:

Verify the divergence theorem for the vectors

a) $\vec{D} = 2\rho z^2\hat{\rho} + \rho\cos^2\phi\hat{z}$ over the region defined by $2 \le \rho \le 5, -1 \le z \le 1, 0 < \phi < \frac{\pi}{3}$.
b) $\vec{A} = r^2\hat{r} + r\sin\theta\cos\phi\hat{\theta}$ over the surface of a quarter of a hemisphere defined by $0 < r < 3, 0 < \phi < \pi/2, 0 < \theta < \pi/2$.

Problem 2.5:

Let $\vec{H} = \rho\sin\phi\hat{\rho} + \rho\cos\phi\hat{\phi} - \rho\hat{z}$

a) Find $\nabla \times \vec{H}$ and $\nabla \times \nabla \times \vec{H}$.
b) Evaluate $\oint_L \vec{H} \cdot \overrightarrow{dl}$ if L is the closed contour of Figure 1 and verify your result using stokes's theorem.

Problem 2.6:

1. Calculate the circulation of the vector field $\vec{B} = -y\hat{x} + x\hat{y} - 25\hat{z}$ around the contour C described by the position vector:

$$\vec{c}(t) = (a_0 + r_0\cos(t))\hat{x} + (b_0 + r_0\sin(t))\hat{y} \text{ where } 0 \le t \le 2\pi.$$

where $a_0, b_0 \epsilon [-\infty, +\infty]$ and $r_0 \epsilon [0, +\infty]$

2. Verify that C is a closed contour, then check your results using Stokes's Theorem.

Problem 2.7:

Given the vector field $\vec{D} = (16xy - z)\hat{x} + 8x^2\hat{y} - x\hat{z}$

(a) Is $\vec{D}$ irrotational (or conservative)?

(b) Find the flux of $\vec{D}$ in the $+y$ direction over the open surface $y = \sin x$ such that $0 < x < \pi$, and $0 < z < 1$

Problem 2.8:

If $\vec{E} = 2\rho \sin\phi\hat{\rho} + 4\rho\cos\phi\hat{\phi} + (z^2 + 1)\rho\hat{z}$,

a) Find $\nabla^2\vec{E}$.

b) Show that $\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2\vec{E}$.

Problem 2.9:

A vector field is given by

$$\vec{Q} = \frac{\sqrt{x^2 + y^2 + z^2}}{\sqrt{x^2 + y^2}}[(x - y)\hat{x} + (x + y)\hat{y}]$$

Consider the shape presented in figure 2, evaluate the following integrals: $\int_L \vec{Q} \cdot \vec{dl}$; $\int_{S_1} (\nabla \times \vec{Q}) \cdot \vec{dS}$; $\int_{S_2} (\nabla \times \vec{Q}) \cdot \vec{dS}$; $\int_{S_1} \vec{Q} \cdot \vec{dS}$; $\int_{S_2} \vec{Q} \cdot \vec{dS}$ and $\int_v \nabla \cdot \vec{Q} dv$|

How do your results in the 1st integral to those in the last integral compare?

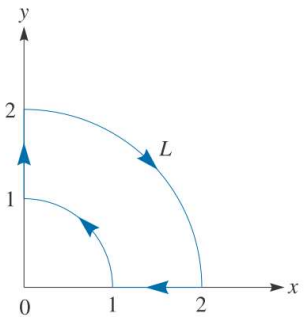


Figure 1

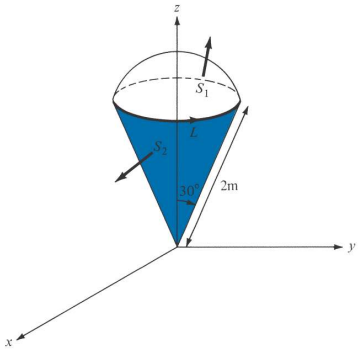


Figure 2